Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

Distance–Time Graphs: Starter

Reading a real graph • two exam questions

Questions: AQA GCSE Physics. Print only, school use.

How to read a distance–time graph

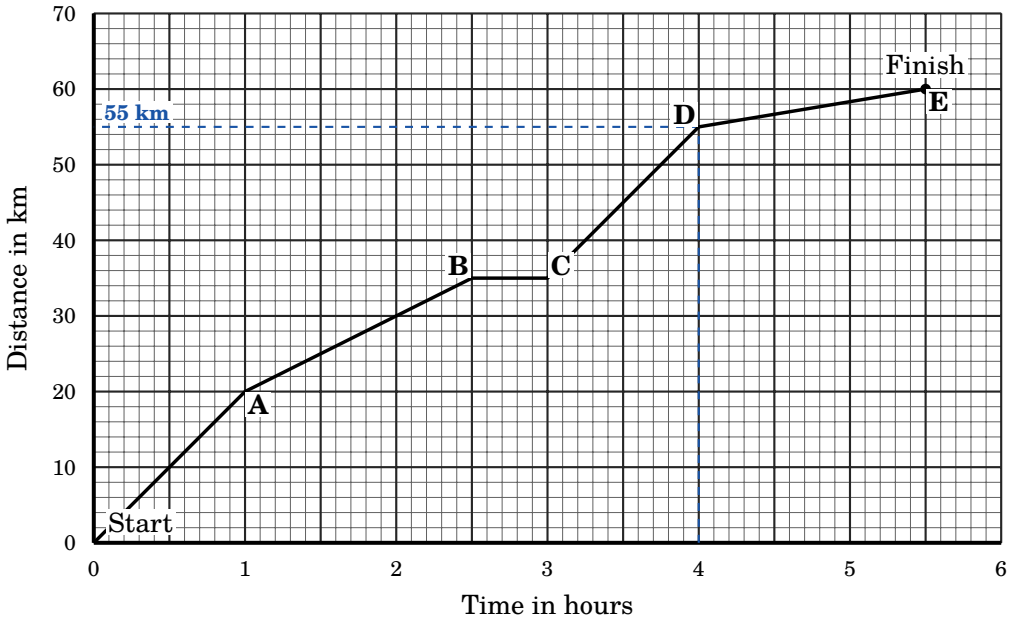
Read any graph in three moves

1. **Read the axes first.** What is up the side, what is along the bottom, and in what units? On the horse graph: distance in **kilometres** up the side, time in **hours** along the bottom.
2. **Read a value with two lines.** Go along the time axis to the time you want, straight **up** to the graph line, then straight **across** to the distance axis. Use a ruler and draw the lines in pencil.
3. **Read the shape.** A **flat** part means stopped. A **steeper** part means faster. A **less steep** part means slower.

Worked example. How far had the horse gone after **4 hours**? Up from 4 on the time axis to the line, then across to the distance axis: **55 km**. The dashed lines on the graph below show it.

Question 1

A horse and rider take part in a long distance race. The graph shows how far the horse and rider travel during the race.



(a) What was the distance of the race?

How to: find *Finish* on the graph and read straight across to the distance axis.

distance = 60 km

[1]

(b) How long did it take the horse and rider to complete the race?

How to: from *Finish*, read straight down to the time axis. Each small square along the bottom is 0.1 hours.

Answer: 5.5 hours (5 hours 30 minutes).

[1]

(c) What distance did the horse and rider travel in the first 2 hours of the race?

How to: draw a line up from **2** on the time axis to the graph, then across, like the worked example.

distance = **30** km

[1]

(d) How long did the horse and rider stop and rest during the race?

How to: resting means the distance stays the same, so find the **flat** part. Read the time at each end of it and take the smaller from the larger.

rest starts at **2.5** h rest ends at **3** h so the rest lasted **0.5 hours**

[1]

(e) Not counting the time it was resting, between which two points was the horse moving the slowest?

How to: the slowest part is the **least steep** sloping part. Fill in the table to compare the parts: how many km, in how many hours.

Part	Start to A	A to B	C to D	D to E
Distance gained / km	**20**	**15**	**20**	**5**
Time taken / h	**1**	**1.5**	**1**	**1.5**

D and **E**

Give a reason for your answer.

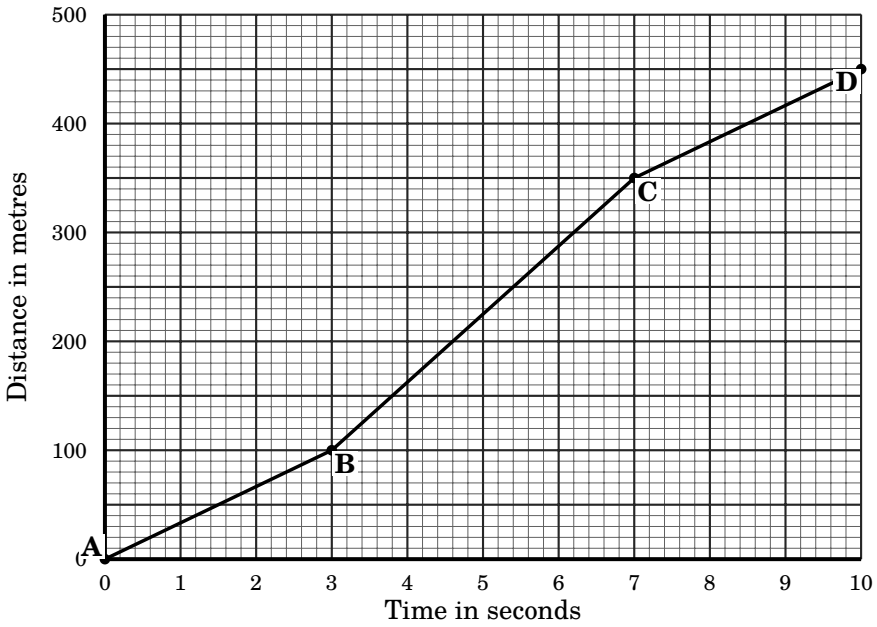
Write it like this

The horse was slowest between **D** and **E**, because the line is **least steep** there: it went only **5** km in **1.5** hours.

[2]

Question 2

The distance–time graph represents the motion of a car during a race.



(a) Describe the motion of the car between point A and point D. You should not carry out any calculations.

How to: take the graph one part at a time. Fill in the table, then turn each row into a sentence.

Part	Straight or curved?	Steeper or less steep than the part before?	So the car is moving at...
A to B	**straight**	*(first part)*	**a constant speed**
B to C	**straight**	**steeper**	**a faster constant speed**
C to D	**straight**	**less steep**	**a slower constant speed**

Write it like this

From A to B the car moves at a ___**constant**___ speed.

From B to C the line is ___**steeper**___, so the car moves at a ___**faster**___ constant speed.

From C to D the line is ___**less steep**___, so the car moves at a ___**slower**___ constant speed.

[3]